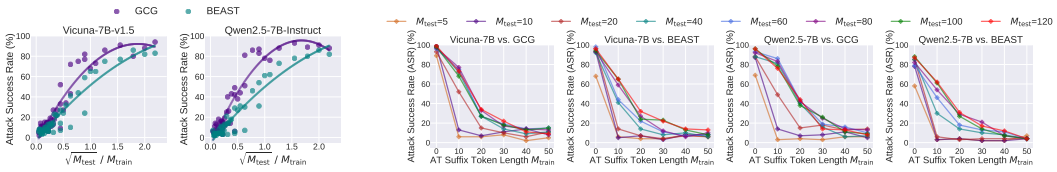
Table 4: ASR (%) of suffix jailbreaking against LLMs trained with Circuit Breakers [73] or LLM AT. A low ASR suggests a strong jailbreak robustness of the targeted model.

(a) ASRs of different jailbreak attacks against Mistral-7B.

Attack	Defense	Adversarial Suffix Token Length M_{test} in Jailbreaking							
		5	10	20	40	60	80	100	120
GCG	Circuit Breakers [72]	21.0	20.0	21.0	23.0	23.0	28.0	28.0	23.0
	LLM AT ($M_{\text{train}} = 20$)	**8.0**	**11.0**	**7.0**	**6.0**	**7.0**	8.0	10.0	11.0
	LLM AT ($M_{\text{train}} = 30$)	11.0	13.0	8.0	**6.0**	**7.0**	**5.0**	**5.0**	**5.0**
BEAST	Circuit Breakers [72]	19.0	21.0	20.0	24.0	25.0	25.0	25.0	27.0
	LLM AT ($M_{\text{train}} = 20$)	**11.0**	**8.0**	**11.0**	**10.0**	**13.0**	**8.0**	**8.0**	**11.0**
	LLM AT ($M_{\text{train}} = 30$)	12.0	13.0	19.0	21.0	18.0	22.0	17.0	22.0

(b) ASRs of different jailbreak attacks against Llama-3-8B.

Model	Defense	Adversarial Suffix Token Length M_{test} in Jailbreaking							
		5	10	20	40	60	80	100	120
GCG	Circuit Breakers [72]	**3.0**	**5.0**	3.0	4.0	3.0	5.0	5.0	7.0
	LLM AT ($M_{\text{train}} = 20$)	5.0	8.0	6.0	5.0	6.0	**1.0**	3.0	1.0
	LLM AT ($M_{\text{train}} = 30$)	11.0	9.0	**2.0**	**3.0**	**0.0**	2.0	**1.0**	**1.0**
BEAST	Circuit Breakers [72]	12.0	**9.0**	11.0	12.0	16.0	15.0	17.0	15.0
	LLM AT ($M_{\text{train}} = 20$)	**10.0**	12.0	**4.0**	13.0	12.0	21.0	19.0	15.0
	LLM AT ($M_{\text{train}} = 30$)	19.0	15.0	12.0	**6.0**	**9.0**	**14.0**	**11.0**	**10.0**



(a) Scatter plots of ASR to the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$.

(b) ASR versus M_{train} under different M_{test}. $M_{\text{train}} = 0$ means that AT is not performed on the evaluated model.

Figure 5: ASR of models trained from the BEAST-based LLM AT. A low ASR indicates a strong jailbreak robustness of the model.

robustness than Circuit Breakers under different jailbreak attack adversarial suffix lengths. (2) When the base model is Llama-3-8B, the two defense methods achieve similar performance.

C.2 LLM AT with the BEAST attack

In our main experiments in Section 5, we solely use the GCG attack to synthesize jailbreak prompts for LLM AT. In this section, we investigate whether our theoretical results still empirically hold for AT with jailbreak attacks other than GCG. Specifically, we now perform LLM AT with the BEAST attack on Vicuna-7B-v1.5 and Qwen2.5-7B-Instruct models. For the hyperparameters of BEAST described in Appendix B.1, we vary the adversarial suffix token length within $\{5, 10, 20, 30, 40, 50\}$, and set k_1 to 64 and k_2 to 16. All other settings of LLM AT follow those described in Section B.2.

Experimental results are presented in Figure 5 and Table 5. From Figure 5a and Table 5, we observe a statistically significant positive correlation between the suffix jailbreak robustness and the ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$ in every experiment, which indicates that our ICL-AT theory still holds for BEAST-based LLM AT. Besides, from Figure 5b, one can find that AT with a short adversarial suffix length M_{train} of 30 can already reduce the ASR from nearly 100% to around 20% in every evaluation case, which demonstrates the effectiveness of short-length BEAST-based LLM AT in defending against jailbreak attacks.

C.3 LLM AT on larger models

We also perform short-length LLM AT on Vicuna-13B-v1.5, which is a model larger than those 7B/8B LLMs adopted in our main experiments in Section 5. All hyperparameters for LLM AT follow those described in Appendix B.2. Results are presented in Table 6, which shows that AT with an

Table 5: PCCs and p-values calculated between ASR and ratio $\sqrt{M_{\text{test}}}/M_{\text{train}}$ on LLMs adversarially trained with the BEAST attack. $p < 5.00 \times 10^{-2}$ means that the correlation between ASR and the ratio is considered statistically significant.

Model	GCG Attack		BEAST Attack	
	PCC(↑)	p-value(↓)	PCC(↑)	p-value(↓)
Vicuna-7B	0.91	$\mathbf{5.3 \times 10^{-19}}$	0.94	$\mathbf{6.7 \times 10^{-24}}$
Qwen2.5-7B	0.88	$\mathbf{2.2 \times 10^{-16}}$	0.95	$\mathbf{5.0 \times 10^{-25}}$

Table 6: ASR (%) of the GCG attack against Vicuna-13B-v1.5 trained with LLM AT. A low ASR suggests a strong jailbreak robustness of the targeted model.

(a) ASRs of different jailbreak attacks against Mistral-7B.

Attack	Defense	Adversarial Suffix Token Length M_{test} in Jailbreaking							
		5	10	20	40	60	80	100	120
GCG	None	92.0	94.0	99.0	96.0	98.0	96.0	99.0	98.0
	LLM AT ($M_{\text{train}} = 5$)	**11.0**	19.0	30.0	53.0	55.0	67.0	70.0	68.0
	LLM AT ($M_{\text{train}} = 20$)	12.0	**9.0**	11.0	**6.0**	**6.0**	**6.0**	**8.0**	**7.0**

adversarial suffix token length as short as 20 can already reduce the ASR of the GCG attack from nearly 99% to around 10% in the worst case. This suggests the generalization of our theoretical findings beyond 7B/8B models.